Answer **one** question from this Section in the spaces provided.

9 **(a)** Define *simple harmonic motion*.

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.... [2]

(b) A flat horizontal plate is made to vibrate in a vertical plane as shown in Fig. 9.1

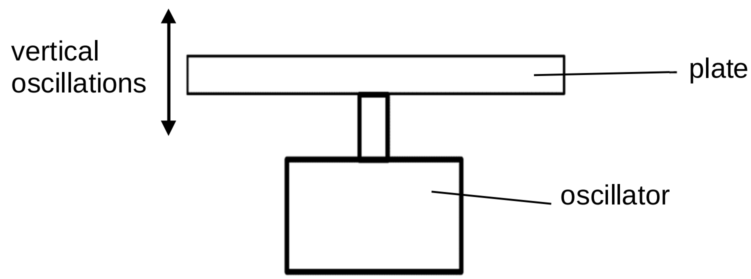

Fig. 9.1

Some sand is sprinkled onto the plate and the variation with displacement x of the acceleration a of the plate is shown in Fig. 9.2. Upward displacement of the plate is taken to be positive. At time t = 0, the plate is at equilibrium position and it is moving upwards.

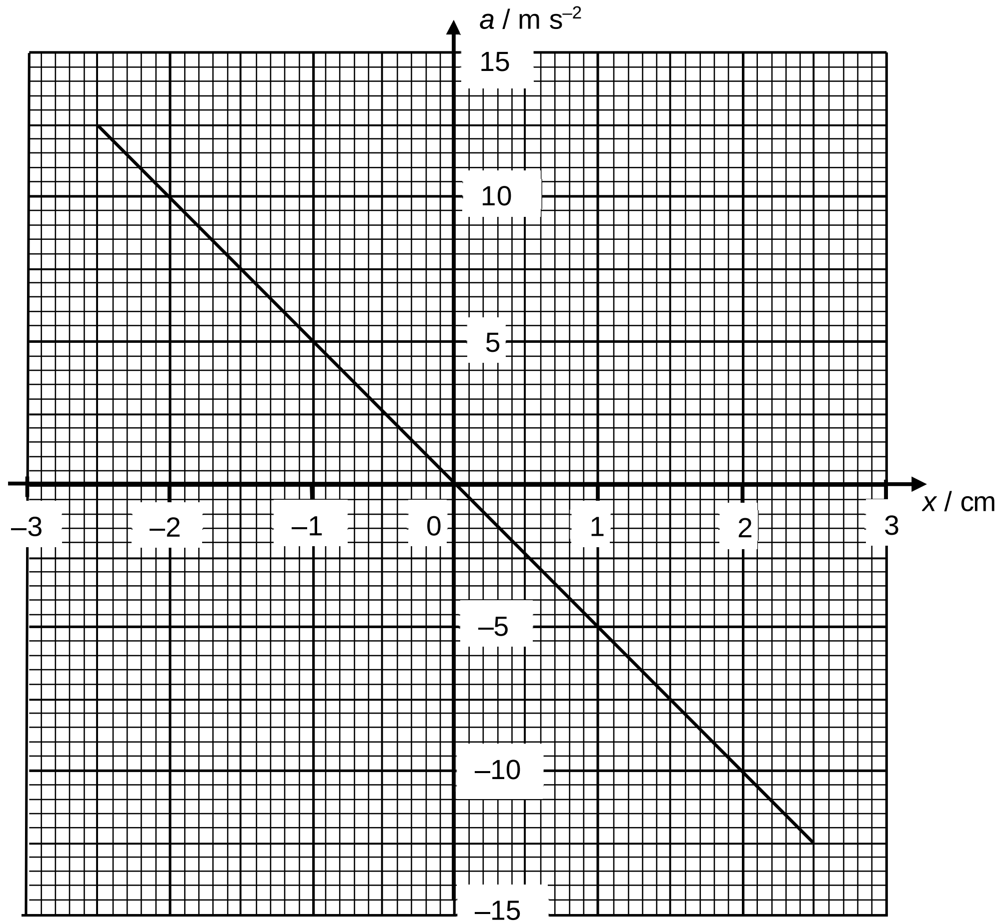

Fig. 9.2

(i) State the acceleration of the plate at which the sand first loses contact with the plate. Explain your reasoning.

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.... [3]

(ii) Assume the mass of each sand particle is 1.2×10^{-5} kg.

Using appropriate values from Fig. 9.2, calculate the

1. frequency of vibration of the oscillation plate,

frequency = ... Hz [3]

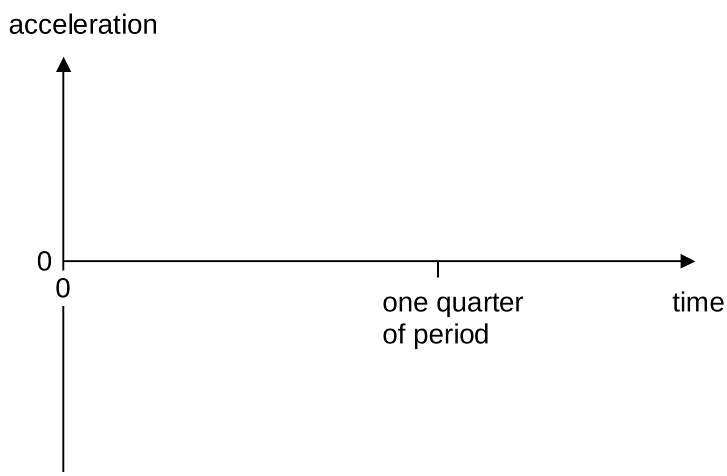
2. kinetic energy of one sand particle at x = – 1.0 cm,

kinetic energy = ... J [2]

3. time when sand particle first loses contact with plate.

time = ... s [2]

(iii) On Fig. 9.3, sketch a graph to show the variation with time of the acceleration of one sand particle for a duration of one quarter of the period of the vibrating plate. Numerical values are not required.



[2]

Fig. 9.3

(c) The bob of a simple pendulum undergoes simple harmonic motion as illustrated in Fig. 9.4.

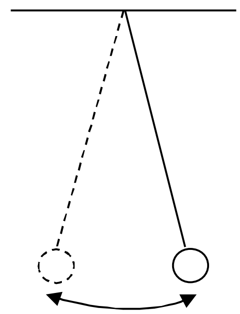


Fig. 9.4

A block of wood, floating in water, undergoes vertical oscillations, as illustrated in Fig. 9.5.

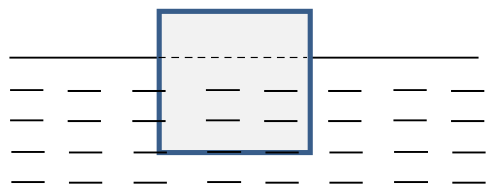


Fig. 9.5

(i) For each system, describe the restoring force that gives rise to the oscillations.

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(ii) For the system in Fig. 9.5, an oscillator is placed in the water and created a wave of variable frequency. At one particular frequency, the block oscillates with a large amplitude.

State the phenomenon as illustrated above and a condition in order for the phenomenon to occur.

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10 (a) A stiff metal wire is used to form a rectangular frame PQRS measuring 8.7 cm by